EE3-18
EE9-AO12

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2018

MSc and EEE PART III/IV: MEng, BEng.and ACGI

MICROWAVE TECHNOLOGY

Tuesday, 4 December 9:00 am

Time allowed: 3:00 hours

Corrected copy

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : O. Sydoruk

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Special instructions for invigilators

This is a closed book examination.

Special instructions for students

All variables and abbreviations have their usual meaning.

Permeability of free space, $\mu_o = 4\pi \times 10^{-7}\ [H/m]$

Permittivity of free space, $\varepsilon_o \approx 8.854\ [pF/m]$

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The Questions

1.

a) Describe the two main contributions to the clear sky noise temperature between 0.1 GHz and 10 GHz.

[3]

b) From 1(a), what is the approximate frequency range that has minimum atmospheric noise. State the ubiquitous application found at such frequencies.

[2]

c) Calculate the approximate wavelength and wavenumber in free space for 1(b).

[1]

d) Explain the principle behind the microwave-based Space Solar Power System (M-SSPS) for wireless energy transmission.

[4]

e) Explain the engineering compromises associated with the proposed frequency of operation for 1(d).

[2]

f) Write down Ampere's law of magnetomotive force.

[2]

g) From 1(f), which term is associated with optics/photonics? At which approximate wavelength and frequency does the performance of optics/photonics begin to drop off?

[2]

h) From 1(f), which term is associated with electronics? At which approximate wavelength and frequency does the performance of electronics begin to drop off?

[2]

i) Briefly comment on the 'THz Gap' and how it is narrowing over time in frequency.

[2]

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2. A coaxial cable has the inner and outer diameters of its dielectric insulator defined by the variables d and D, respectively. The RG-6/U coaxial cable has $d = 1.016$ mm, $D = 4.572$ mm, with the insulator dielectric constant $\varepsilon_r = 1.487$ and room-temperature resistivity $\rho_o = 1.68 \times 10^{-8}\ \Omega.\text{m}$ for the copper conductors.

a) Using the over-simplified linear strip approach, derive the total inductance per unit length in terms of classical skin depth. In addition, derive the parallel-plate capacitance per unit length for the average circumference of the insulator and the associated characteristic impedance.

[4]

b) From 2(a), for the RG-6/U cable, calculate the classical skin depth at 25 GHz and associated inductance per unit length. In addition, calculate the capacitance per unit length and the associated characteristic impedance.

[4]

c) Exact expressions for the respective inductance and capacitance per unit length of a coaxial cable are given by:

$$L' = \frac{\mu}{2\pi} ln\left\{\frac{D}{d}\right\} \qquad \text{[H/m]} \qquad (2.1)$$

$$C' = \frac{2\pi\varepsilon}{ln\left\{\frac{D}{d}\right\}} \qquad \text{[F/m]} \qquad (2.2)$$

For the RG-6/U coaxial cable, calculate the inductance and capacitance per unit length and associated characteristic impedance.

[2]

d) From the calculated characteristic impedance in 2(c), what kind of application does this particular type of coaxial cable have and why?

[2]

e) Comment on the frequency dispersive nature of the exact expressions, when compared to those derived in 2(a), and the calculated results for the RG-6/U cable. What conclusion can be made from the use of the over-simplified linear strip approach?

[2]

f) With a coaxial cable, what is the dominant mode of propagation and its associated lower cut-off frequency?

[2]

g) The mode with the next lowest cut-off frequency is the TE_{11} mode. To a good approximation, the condition for the TE_{11} mode to propagate is that the wavelength in the dielectric is no longer than the average circumference of the insulator. Derive an expression for the cut-off frequency for the TE_{11} mode.

[2]

h) Using 2(g), for the RG-6/U coaxial cable, calculate the lower cut-off frequency for the TE_{11} mode and give a reasonable upper cut-off frequency for the dominant mode.

[2]

3.

a) A stripline and microstrip transmission lines, having the same characteristic impedance, are to be implemented for the same total board thickness.

i) Sketch their cross-sections and comment on the relative frequencies of their transvers resonant modes.

[2]

ii) What is the difference in their dominant mode of propagation when the dielectric constant of the insulator is increased from the free-space value?

[2]

iii) State one prominent advantage for stripline and one for microstrip.

[2]

b) At a distance L_1 from the input port, the track width for a lossless 50 Ω stripline is abruptly changed to give 35.355 Ω, with the ends of each line terminated by matched port impedances of 50 Ω and 35.355 Ω, respectively.

i) Beyond the distance L_1, does the track width increase or decrease? As a result, comment of the changes in the inductance per unit length and capacitance per unit length for this 35.355 Ω transmission line.

[2]

ii) Calculate the incident voltage-wave reflection coefficient at the discontinuity. At the input port, how does the incident voltage-wave reflection coefficient change with frequency?

[2]

c) With the scenario in 3(b), the output port impedance is now increased to 50 Ω.

i) At the input port, what is the incident voltage-wave reflection coefficient at zero frequency?

[1]

ii) At the input port, constructive interference will be seen at frequency f_1. At this frequency, what will be the electrical length for the 35.355 Ω transmission line?

[2]

iii) At the input port, destructive interference will be seen at frequency f_2. At this frequency, what will be the electrical length for the 35.355 Ω transmission line?

[2]

iv) Sketch the frequency response for the magnitude of the incident voltage-wave reflection coefficient as seen at the input port.

[1]

v) Derive an expression for the difference frequency $\Delta f = (f_2 - f_1)$ in terms of the physical length of the 35.355 Ω transmission line L_2.

[3]

vi) With $\Delta f = 1$ GHz and the dielectric constant of the insulator being $\varepsilon_r = 2.45$, calculate L_2.

[1]

4. This question relates to a lossless air-filled metal-pipe rectangular waveguide operating in its dominant transverse electric TE_{10} mode of propagation.

a) Using the generic relationship for propagation constant γ as a starting point, given by equation (4.1), derive the expression for guided wavelength λ_g in terms of the free-space wavelength λ_o and cut-off wavelength λ_c.

$$\gamma_o^2 = \gamma_x^2 + \gamma_y^2 + \gamma_z^2 \tag{4.1}$$

[4]

b) What is the relationship between the cut-off wavelength for the dominant TE_{10} mode and the internal broad dimension a?

[1]

c) Using the expression derived in 4(a), what happens to the propagation constant below the cut-off frequency?

[2]

d) Using the generic relationship between impedance η and propagation constant, given by equation (4.2), derive the wave impedance Z_{TE} in terms of the intrinsic impedance of free space η_o, free-space wavelength λ_o and cut-off wavelength λ_c.

$$\eta = \frac{j\omega\mu}{\gamma} \tag{4.2}$$

[2]

e) The WR-90 waveguide has internal broad dimensions of $a = 22.86$ mm. At an operating frequency $f_o = 1.5\, f_c$, where f_c is the cut-off frequency, calculate:

i) Cut-off frequency.

[1]

ii) Guided wavelength.

[2]

iii) Physical length for the TE_{101} box mode.

[1]

iv) Wave impedance.

[2]

f) The slotted-line waveguide experiment can directly measure the magnitude of the standing wave $|V|$. Briefly explain the physical characteristics of the slot (e.g., its shape and position) and the diode detector's *I-V* characteristics that allow such measurements to be undertaken.

[2]

g) The relationship between the magnitude of the voltage-wave reflection coefficient $|\rho|$ of the impedance terminating the end of the waveguide and slotted-line measured voltage standing wave ratio $VSWR = |V|_{MAX}/|V|_{MIN}$ is given by equation (4.3).

$$|\rho| = \frac{VSWR-1}{VSWR+1} \tag{4.3}$$

Determine the *VSWR* when the waveguide is terminated by an open circuit at an operating frequency $f_o = 1.5\, f_c$, where f_c is the cut-off frequency.

[3]

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5.

a) For a parallel RLC lumped-element resonator:

i) State the fundamental equation for quality factor in terms of energy.
[1]

ii) State the energy equation associated with the resistor.
[1]

iii) State the energy equation associated with the capacitor.
[1]

iv) Using 5(a)(i-iii), state the resulting equation for quality factor.
[1]

v) State the frequency equation associated with 5(a)(iv), in terms of lumped-elements, and define its damping/driven state.
[2]

b) The eigenfrequencies for an RLC resonator are given by equation (5.1).

$$\tilde{\omega}_{0,1} = \pm\omega_o\sqrt{1-\frac{1}{(2Q_u(\omega_o))^2}} + j\frac{\omega_o}{2Q_u(\omega_o)} \qquad (5.1)$$

i) From equation (5.1), what is the damped (or undriven) natural angular resonance frequency?
[1]

ii) From equation (5.1), what is the amplitude decay rate or loss rate or Napier frequency?
[1]

iii) What is the amplitude decay time constant?
[1]

iv) Clearly sketch the voltage waveform, as a function of time, for a driven and undriven tank resonator. Show all related variables.
[2]

v) Using equation (5.1), derive an expression that relate quality factors at the driven and undriven resonance frequencies.
[2]

vi) With critical damping, what are the values of quality factors at the driven and undriven resonance frequencies?
[1]

c) For a parallel RLC lumped-element resonator, R = 100 Ω, L = 5.305 nH and C = 6.366 pF, calculate:

i) Undamped resonance frequency.
[1]

ii) Quality factor at the undamped resonance frequency.
[1]

iii) Damped resonance frequency.
[1]

iv) Quality factor at the damped resonance frequency.
[1]

v) Amplitude decay time constant.
[1]

vi) -3 dB bandwidth.
[1]

6. a) Describe, with the aid of a diagram, a branch-line coupler having a distributed-element implementation, indicating lengths and impedances. Indicate its main characteristics.

[5]

b) Replace the distributed-element implementation in 6(a) with an equivalent lumped-element version. How does the performance compare with that in 6(a)? With a coupler having an impedance of $Zo = 50\ \Omega$, calculate the components values for a design frequency of 1.85 GHz.

[5]

c) Replace the lumped-element implementation in 6(b) with an equivalent lumped-distributed version. With a coupler having an impedance of $Zo = 50\ \Omega$ and line sections of electrical length $\phi = 45°$, calculate the components values for a design frequency of 1.85 GHz.

[5]

d) Draw the topology of a balanced amplifier employing the coupler given in 6(a).

[5]